

Spoken Digit Recognition from Short Audio Clips Using Mel-Spectrogram Features and ResNet-18

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Abstract—This paper presents a spoken digit recognition system for classifying audio recordings of digits 0–9 under challenging conditions including unseen speakers and noisy recordings. Raw audio is converted into 3-channel Mel-spectrogram representations incorporating delta and delta-delta features, classified using a modified ResNet-18 trained from scratch. A multi-level augmentation strategy at both waveform and spectrogram levels improves cross-speaker generalization. Training uses 5-fold stratified cross-validation with test-time augmentation and fold ensembling at inference. The system achieves an out-of-fold validation accuracy of 99.54% on the competition dataset, evaluated using macro-averaged F1-score over all ten digit classes.

Index Terms—spoken digit recognition, Mel-spectrogram, ResNet-18, SpecAugment, Mixup, cross-validation, audio classification

I. INTRODUCTION

Automatic recognition of spoken digits has direct applications in telephone banking, PIN entry systems, and voice-controlled embedded devices. While modern end-to-end systems achieve high accuracy in controlled settings, performance degrades when evaluation speakers are absent from training or recordings are noisy. This work addresses precisely that generalization challenge.

The task is a 10-class classification problem: predict the spoken digit (0–9) from a short .wav clip. The test split contains unseen speakers and recordings with additive noise or temporal artifacts. A system that memorizes speaker-specific patterns will fail to transfer. The objective is to learn digit-discriminative representations that are robust to speaker identity and recording variability.

We adopt an image-classification framing: each audio clip is transformed into a 2D time-frequency image and classified with a CNN. This is well-motivated since short utterances encode digit identity primarily in their spectro-temporal shape, which CNNs are well-suited to capture, and it enables direct use of powerful spectrogram-domain augmentation to simulate test-time acoustic variation.

II. PREPROCESSING AND FEATURE EXTRACTION

A. Audio Normalization

Audio files are loaded as mono waveforms and resampled to 16 kHz. Each clip is standardized to exactly 1 second (16,000 samples): shorter clips are center-padded with zeros and longer clips are center-trimmed. This ensures all inputs share the same shape and removes any positional bias arising from variable recording durations.

B. Mel-Spectrogram Computation

The waveform is converted to a log-Mel spectrogram with $n_{\text{mels}} = 128$, $n_{\text{fft}} = 512$, $\text{hop_length} = 160$, over a frequency range of 50–8000 Hz. The power spectrogram is converted to decibel scale to compress the dynamic range and improve numerical stability during training. The resulting spectrogram has dimensions 128×101 (frequency bins $\times$ time frames).

C. Delta Features and Input Stacking

First-order (delta) and second-order (delta-delta) features are computed via `librosa.feature.delta` [4] and stacked with the base log-Mel spectrogram along the channel axis to form a $3 \times 128 \times 101$ tensor, analogous to a 3-channel RGB image. The delta encodes the rate of change of the Mel spectrum over time, while the delta-delta captures its acceleration. This 3-channel representation gives the model both static spectral content and temporal dynamics, both essential for distinguishing phonetically similar digit classes across diverse speakers.

D. Precomputation

To eliminate repeated spectrogram extraction during training, all train and test spectrograms are computed once offline and stored as .npy binary files. The DataLoader reads directly from disk at runtime, bypassing `librosa` entirely. This reduced per-epoch training time from over 15 minutes to under 3 minutes, making 5-fold, 35-epoch training feasible within Kaggle GPU session limits.

III. DATA AUGMENTATION

Generalizing to unseen speakers requires invariance to speaker-specific acoustic factors irrelevant to digit identity. Two levels of augmentation are applied during training to achieve this.

Waveform-level augmentation is applied to the raw audio before spectrogram conversion. Additive white Gaussian noise with amplitude sampled from $[0.002, 0.008]$ is added with probability 0.5, simulating microphone noise and degraded recording conditions. A random circular time shift of up to ± 0.1 s is also applied, displacing the utterance within the 1-second window and preventing reliance on absolute speech onset position.

Spectrogram-level augmentation follows the SpecAugment policy [1]. Frequency masking zeros out 2 randomly chosen bands of up to 16 consecutive Mel bins each, and time

masking zeros out 2 intervals of up to 30 consecutive time frames each. These masks force the model to recognize digits from partial spectral and temporal information, improving robustness to real-world recording degradation.

Mixup [2] is applied with probability 0.5 and mixing coefficient $\lambda \sim \text{Beta}(0.4, 0.4)$. Training pairs are blended as $\tilde{x} = \lambda x_i + (1-\lambda)x_j$ and $\tilde{y} = \lambda y_i + (1-\lambda)y_j$, smoothing decision boundaries and acting as a strong regularizer against overconfident predictions. Label smoothing with $\epsilon = 0.1$ is additionally applied in the cross-entropy loss function.

IV. MODEL ARCHITECTURE

The backbone is ResNet-18 [3], a 18-layer residual convolutional network trained entirely from scratch on the spectrogram domain without pretrained weights, as the statistical structure of audio spectrograms differs substantially from natural images.

Three targeted modifications adapt the standard architecture to our task. First, `conv1` is modified to accept 3-channel inputs (log-Mel, delta, delta-delta) instead of RGB. Second, the initial max-pooling layer following `conv1` is replaced with `nn.Identity()` to preserve fine-grained spatial resolution—important because the 128×101 spectrogram is considerably smaller than the 224×224 images ResNet-18 was originally designed for, and aggressive early downsampling would discard critical time-frequency detail. Third, the final fully connected layer is replaced with a custom classification head:

$$\text{Drop}(0.2) \rightarrow \text{Lin}(512 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Drop}(0.1) \rightarrow \text{Lin}(256 \rightarrow 10) \rightarrow \text{Logit} \quad (1)$$

The dropout layers provide additional regularization in the high-dimensional feature space. Total trainable parameters: ≈ 11.2 million. The output is a 10-dimensional logit vector over digit classes 0–9.

V. TRAINING SETUP

Training is performed under 5-fold stratified K-Fold cross-validation, which partitions labeled data into five folds while preserving class distribution in each fold, ensuring balanced digit representation across all validation splits. The AdamW optimizer [5] is used with learning rate 3×10^{-4} and weight decay 10^{-4} . The learning rate is annealed using a cosine schedule (`CosineAnnealingLR`, $T_{\max} = 35$, $\eta_{\min} = 10^{-6}$) for smooth convergence. Each fold runs for 35 epochs with batch size 256. Mixed-precision training (`torch.cuda.amp`) reduces GPU memory consumption and accelerates computation. After each epoch, validation loss and accuracy are evaluated on the held-out fold, and the best checkpoint is saved.

VI. INFERENCE

At inference, test-time augmentation (TTA) is applied to reduce prediction variance: each test spectrogram is independently augmented 5 times using the training augmentation pipeline, and the resulting softmax probability vectors are

averaged element-wise. The final prediction is obtained by averaging the TTA-averaged outputs across all 5 fold models, yielding a 25-forward-pass ensemble per test sample. Ensembling in probability space before the final argmax ensures that consistent, high-confidence predictions are reinforced while uncertain predictions across folds are dampened.

VII. RESULTS AND DISCUSSION

Table I reports the best validation accuracy per fold and the overall out-of-fold (OOF) accuracy aggregated across all five folds.

TABLE I
PER-FOLD BEST VALIDATION ACCURACY

Fold	Best Validation Accuracy
Fold 1	99.55%
Fold 2	99.42%
Fold 3	99.56%
Fold 4	99.63%
Fold 5	99.54%
OOF (Overall)	99.54%

Convergence was rapid: validation accuracy exceeded 99% by approximately epoch 8 in most folds, and training loss decreased steadily from ≈ 1.54 to ≈ 0.69 over 35 epochs. The high consistency across folds (std $< 0.08\%$) confirms that the model is stable and the training procedure is insensitive to the specific data partition used. Predictions were generated for all 16,200 test samples and submitted to the competition leaderboard.

The strong performance reflects several design decisions working in concert. The 3-channel spectrogram encoding was a critical choice: delta and delta-delta features supply temporal context that a static log-Mel representation alone cannot capture, which is particularly useful for discriminating digits with phonetically similar vowel sounds across speaker accents. The multi-level augmentation stack—especially the combination of Mixup and spectrogram masking—was the primary driver of cross-speaker generalization, as digit-specific acoustic patterns are easily obscured by speaker variability without explicit regularization. Removing the initial max-pooling layer additionally allowed the network to retain fine time-frequency structure that would otherwise be discarded in early downsampling.

VIII. CONCLUSION

A spoken digit recognition pipeline was presented using 3-channel Mel-spectrogram features and a modified ResNet-18 trained from scratch. Multi-level waveform and spectrogram augmentation, Mixup regularization, 5-fold stratified cross-validation, and TTA-based ensemble inference together achieved 99.54% OOF validation accuracy on a dataset with unseen speakers and noisy recordings. These results confirm that compact, well-regularized convolutional models with carefully engineered input representations and augmentation strategies can achieve near-perfect accuracy on narrow audio classification tasks, without requiring large pretrained models or complex sequential architectures.

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